

1.1.5a/b/c

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1.1.5a

$$1. \quad 3.5V = 470\Omega \cdot I \quad I = 0.00744A = 7.44mA$$

a.

$$b. \quad 3.5V = 15.54A \cdot R \quad R = 226k\Omega$$

$$c. \quad V = 1.4mA \cdot 22k\Omega \quad V = 30.8V$$

$$d. \quad 2.4mV = 104A \cdot R \quad R = 240\Omega$$

$$e. \quad 20V = 6.8k\Omega \cdot I \quad I = 2.94mA$$

$$f. \quad V = 2.5A \cdot 120\Omega \quad V = 300V$$

$$2. \quad 820 + 1200 + 150 = 2170\Omega$$

$$b. \quad 470 + 3900 + 1800 + 4700 = 10870\Omega$$

$$c. \quad \frac{1}{R_T} = \frac{1}{5600} + \frac{1}{680} + \frac{1}{8200} = 565\Omega$$

$$d. \quad \frac{1}{R_T} = \frac{1}{600} + \frac{1}{200} + \frac{1}{600} + \frac{1}{300} + \frac{1}{200} = 66.7\Omega$$

$$3. \quad \text{Series so } R_T = 470 + 1200 + 270 = 1940\Omega$$

$$V_T = I_T R_T, \quad I_T = \frac{9}{1940} = 4.64mA$$

$$V_1 = I_T R_1 = 2.18V \quad V_2 = I_T R_2 = 5.57V \quad V_3 = I_T R_3 = 1.25V$$

$$4. \quad \text{Series } R_T = 2.7k + 10k + 3.3k + 8.2k = 24.2k\Omega$$

$$I_T = V_T / R_T = 248\mu A \quad V_1 = I_T R_1 = 2.7k\Omega \cdot 248\mu A = 0.669V$$

$$V_2 = I_T R_2 = 248\mu A \cdot 10k\Omega = 2.48V \quad V_3 = 248\mu A \cdot 3.3k\Omega = 0.818V$$

$$V_4 = 248\mu A \cdot 8200\Omega = 2.03 \quad 2.03 + 2.48 + 0.818 + 0.669 = 6V$$

$$5. \quad \frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{8.2k} + \frac{1}{6.8k} + \frac{1}{4.7k} = 2075\Omega$$

$$I_T = V_T / R_T = 5.78mA \quad I_1 = V_T / R_1 = 1.46mA$$

$$V_T = 12V \quad I_2 = V_T / R_2 = 1.76mA \quad I_3 = V_T / R_3 = 2.55mA$$

$$1.76 + 1.46 + 2.55 = 5.77mA$$

$$6. \quad V_T = 6V \quad \frac{1}{R_T} = \frac{1}{2700} + \frac{1}{10000} + \frac{1}{2300} + \frac{1}{8200} = 1117\Omega$$

$$I_T = V_T / R_T = 5.37mA \quad I_1 = V_T / R_1 = 2.22mA$$

$$I_2 = V_T / R_2 = 0.6mA \quad I_3 = V_T / R_3 = 1.3mA \quad I_4 = V_T / R_4 = 0.73mA$$

$$7. \quad R_2 + R_3 = R_{23} = 670\Omega \quad R_4 \text{ in parallel w/ } R_{23} \quad \frac{670 \cdot 1800}{1800 + 670} = 488.26\Omega$$

$$488.26 + 120 + 270 = 908.26\Omega \quad R_T$$

$$V_T = I_T R_T \quad I_T = V_T / R_T = 12V / 908.26 = 13.21mA$$

1.1.5b

I: 17.9 mA
I(p-p): 0 A
I(rms): 0 A
I(dc): 17.9 mA
I(freq): --

V: 0 V
V(p-p): 0 V
V(rms): 0 V
V(dc): 0 V
V(freq): --

V: 8.42 V
V(p-p): 0 V
V(rms): 0 V
V(dc): 8.42 V
V(freq): --

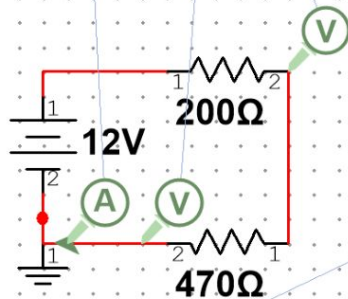
I: 248 uA
I(p-p): 0 A
I(rms): 0 A
I(dc): 248 uA
I(freq): --

V: 5.33 V
V(p-p): 0 V
V(rms): 0 V
V(dc): 5.33 V
V(freq): --

V: 2.85 V
V(p-p): 0 V
V(rms): 0 V
V(dc): 2.85 V
V(freq): --

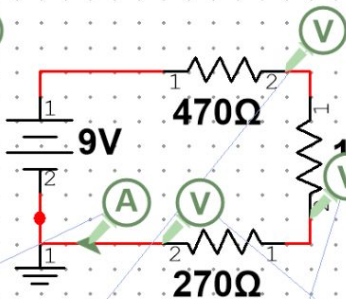
V: 2.03 V
V(p-p): 0 V
V(rms): 0 V
V(dc): 2.03 V
V(freq): --

V: 0 V
V(p-p): 0 V
V(rms): 0 V
V(dc): 0 V
V(freq): --



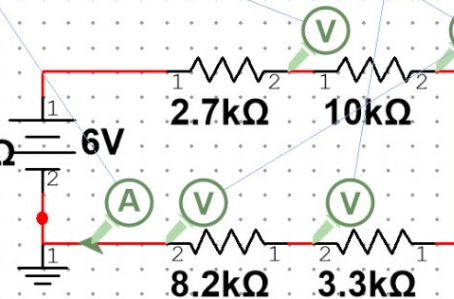
I: 4.64 mA
I(p-p): 0 A
I(rms): 0 A
I(dc): 4.64 mA
I(freq): --

V: 6.82 V
V(p-p): 0 V
V(rms): 0 V
V(dc): 6.82 V
V(freq): --

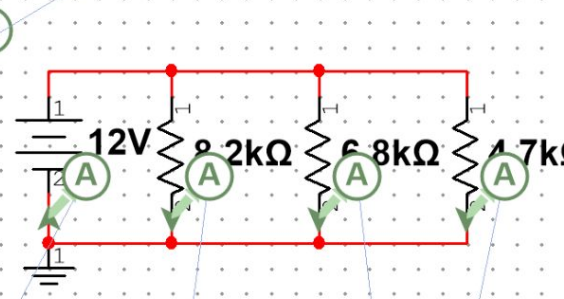


V: 1.25 V
V(p-p): 0 V
V(rms): 0 V
V(dc): 1.25 V
V(freq): --

V: 0 V
V(p-p): 0 V
V(rms): 0 V
V(dc): 0 V
V(freq): --



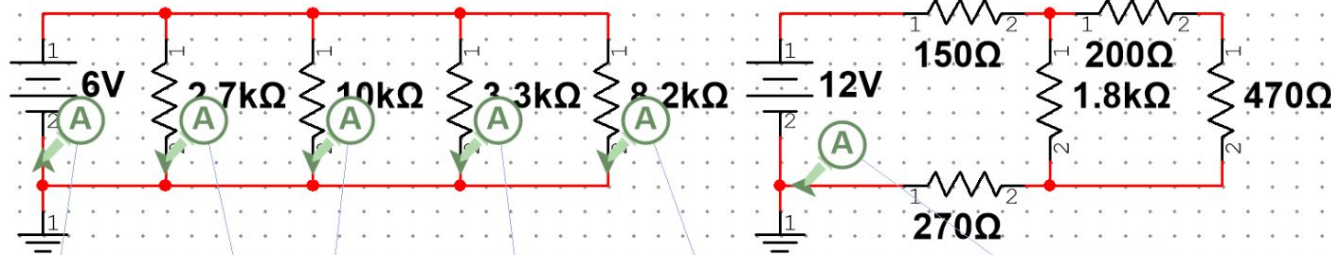
I: 5.78 mA
I(p-p): 0 A
I(rms): 0 A
I(dc): 5.78 mA
I(freq): --



I: 1.46 mA
I(p-p): 0 A
I(rms): 0 A
I(dc): 1.46 mA
I(freq): --

I: 1.76 mA
I(p-p): 0 A
I(rms): 0 A
I(dc): 1.76 mA
I(freq): --

I: 2.55 mA
I(p-p): 0 A
I(rms): 0 A
I(dc): 2.55 mA
I(freq): --



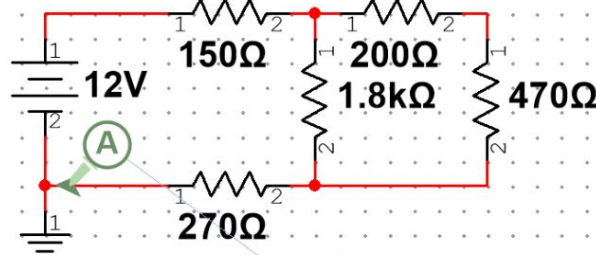
I: 5.37 mA
I(p-p): 0 A
I(rms): 0 A
I(dc): 5.37 mA
I(freq): --

I: 2.22 mA
I(p-p): 0 A
I(rms): 0 A
I(dc): 2.22 mA
I(freq): --

I: 600 uA
I(p-p): 0 A
I(rms): 0 A
I(dc): 600 uA
I(freq): --

I: 1.82 mA
I(p-p): 0 A
I(rms): 0 A
I(dc): 1.82 mA
I(freq): --

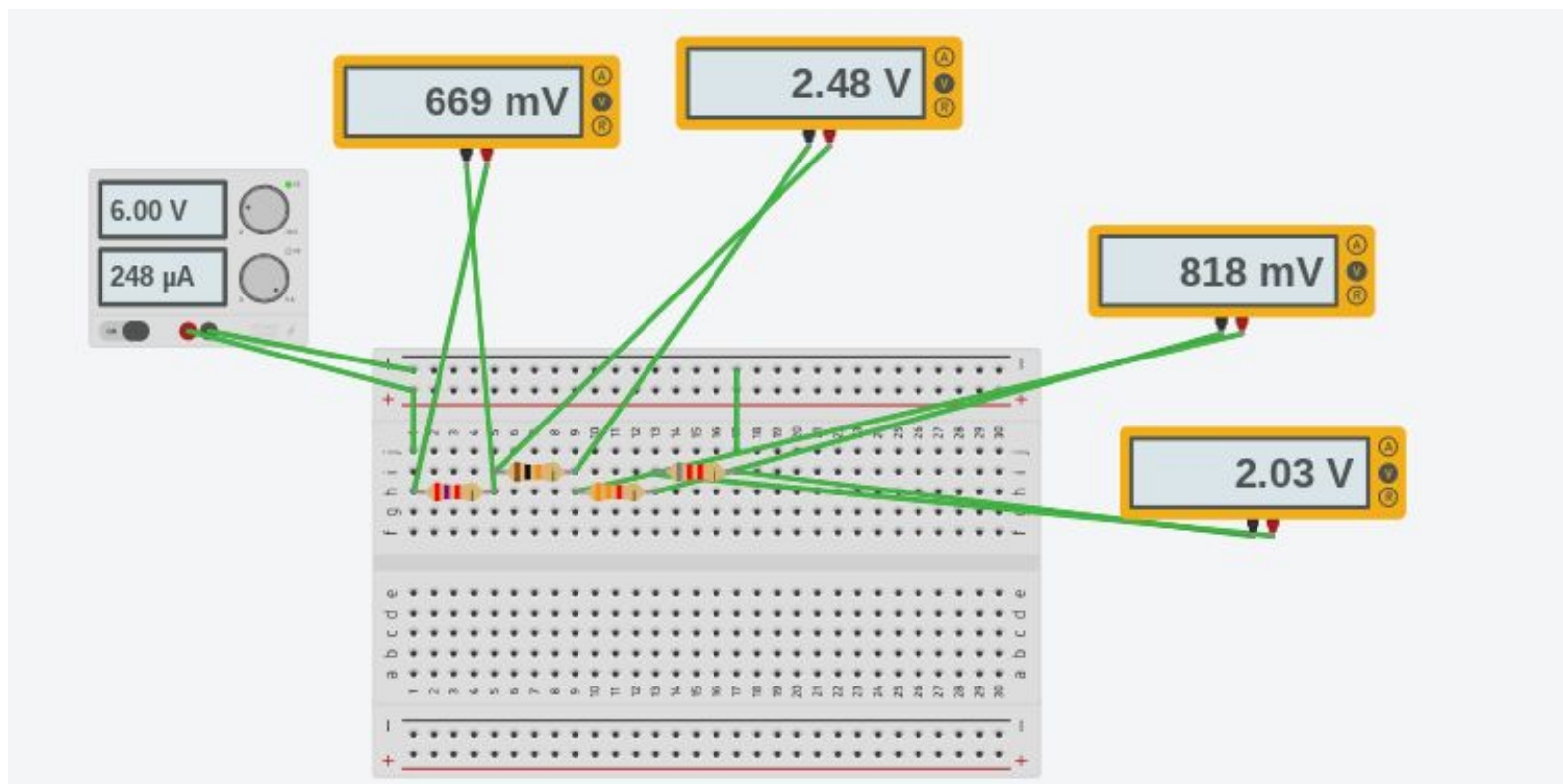
I: 732 uA
I(p-p): 0 A
I(rms): 0 A
I(dc): 732 uA
I(freq): --

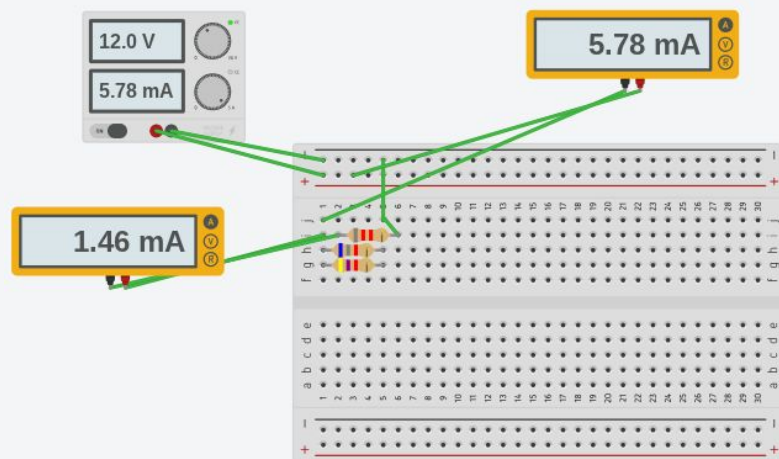
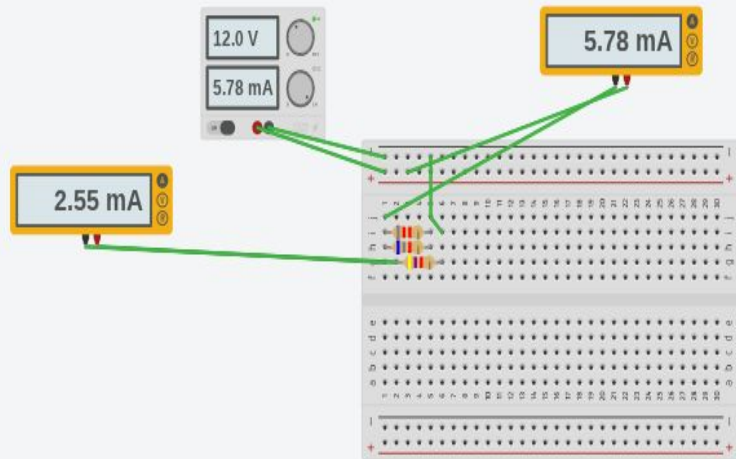


I: 13.2 mA
I(p-p): 0 A
I(rms): 0 A
I(dc): 13.2 mA
I(freq): --

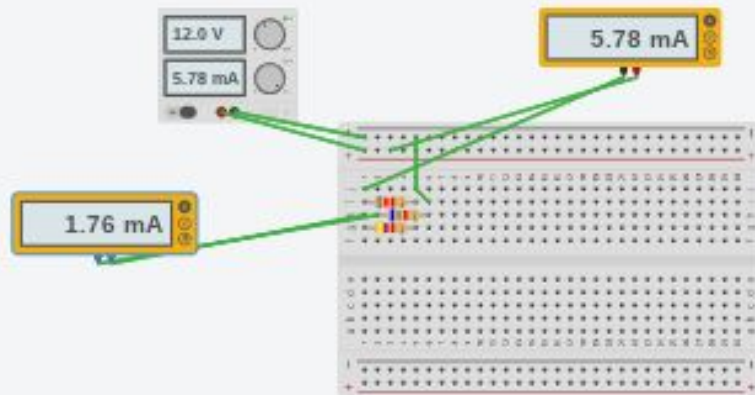
1.1.5c

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